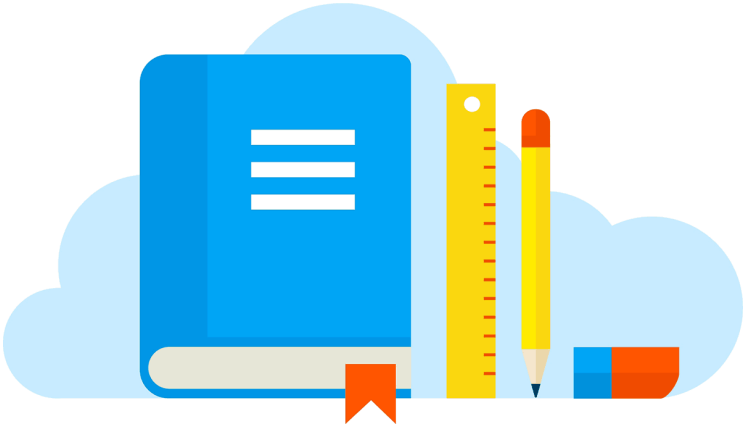


📘 NOTE 4 — DBMS FUNDAMENTALS

Date: 11/04/2026

DBMS FUNDAMENTALS

Database Management System Notes

Subject: DBMS
Topic: DBMS Fundamentals
Level: BCA Semester 5

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1. Introduction to DBMS

DBMS stands for Database Management System.

A DBMS is software used to **create, store, organize, update, and retrieve data** from a database.

Instead of storing information in separate files, a DBMS allows data to be stored in an organized and structured manner.

Examples of DBMS

- MySQL
- Oracle
- PostgreSQL
- Microsoft SQL Server
- SQLite

In **StudySphere**, a database can store information about students, notes, assignments, submissions, and grades.

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2. What is a Database?

A **database** is an organized collection of related data.

For example, a student database may contain:

Student ID

Student Name

Email

Course

Semester

An assignment database may contain:

Assignment ID

Title

Subject

Description

Due Date

The DBMS provides tools to manage this information efficiently.

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3. Need for DBMS

A DBMS is required because managing large amounts of data manually can be difficult.

Main reasons for using DBMS:

- Easy data storage
- Fast data retrieval
- Data security
- Reduced data duplication
- Data consistency
- Backup and recovery
- Multiple users can access data
- Easy data modification

For example, in StudySphere, an admin can add a new note to the database and students can retrieve that note when they open the Notes page.

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4. Components of DBMS

A database system generally consists of:

1. Data

The actual information stored in the database.

2. DBMS Software

Software used to manage the database.

3. Users

People who interact with the database.

Examples:

- Admin
- Student
- Database Administrator

4. Hardware

Computers and storage devices used to run and store the database.

5. Procedures

Rules and instructions used to manage the database.

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5. DBMS Architecture

A simple database system can be represented as:

User

↓

Application

↓

DBMS

↓

Database

For example, in StudySphere:

Student

↓

StudySphere Website

↓

Backend

↓

MySQL Database

When a student logs in, the application sends a request to the backend. The backend communicates with the database and returns the required information.

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6. Tables

A **table** is used to store data in rows and columns.

Example:

Student Table

S	N	Email

1	A	aarav@example.com
1	R	rohan@example.com

Row

A row represents one complete record.

Column

A column represents a particular type of information.

For example, `Name` is a column and `Aarav Mehta` is a value in that column.

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7. Keys in DBMS

Keys are used to identify records and establish relationships between tables.

Primary Key

A **Primary Key** uniquely identifies each record in a table.

Example:

Student_ID

Two students cannot have the same Student_ID.

Foreign Key

A **Foreign Key** is used to create a relationship between two tables.

For example:

Student Table

Student_ID

↓

Submission Table

Student_ID

The `Student_ID` in the Submission table can refer to the student in the Student table.

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8. SQL

SQL stands for Structured Query Language.

SQL is used to communicate with relational databases.

It can be used to:

- Create tables
- Insert data
- Retrieve data
- Update data
- Delete data

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9. Basic SQL Commands

CREATE

Used to create a database or table.

```
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100)
);
```

INSERT

Used to add data.

```
INSERT INTO Students
VALUES (101, 'Aarav Mehta', 'aarav@example.com');
```

SELECT

Used to retrieve data.

```
SELECT * FROM Students;
```

UPDATE

Used to modify existing data.

```
UPDATE Students
SET name = 'Aarav Sharma'
WHERE student_id = 101;
```

DELETE

Used to remove data.

```
DELETE FROM Students
WHERE student_id = 101;
```

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10. DBMS Relationships

Relationships describe how tables are connected.

One-to-One

One record in one table is related to one record in another table.

One-to-Many

One record can be related to many records.

Example:

```
One Student
   ↓
Many Submissions
```

Many-to-Many

Many records in one table can be related to many records in another table.

Example:

Students ↔ Subjects

A student can study many subjects, and a subject can have many students.

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11. Data Integrity

Data integrity means maintaining the accuracy and consistency of data in a database.

DBMS uses constraints to maintain data integrity.

Common constraints include:

- `PRIMARY KEY`
- `FOREIGN KEY`
- `NOT NULL`
- `UNIQUE`
- `CHECK`
- `DEFAULT`

Example:

email VARCHAR(100) UNIQUE

This prevents duplicate email addresses from being stored.

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12. DBMS Security

Database security protects data from unauthorized access.

Common security methods include:

- Username and password authentication
- User permissions
- Access control
- Data encryption
- Regular backups

For example, in StudySphere, an admin should have permission to manage notes and assignments, while a student should mainly access learning materials and submit assignments.

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13. DBMS in StudySphere

The database is an important part of the StudySphere project.

It can store:

Users

↓

Notes

↓

Assignments

↓

Submissions

↓

Grades

For example:

Users table

Stores student and admin account information.

Notes table

Stores note title, subject, description, and file information.

Assignments table

Stores assignment title, subject, instructions, and due date.

Submissions table

Stores student submissions and their status.

Grades

Stores marks or grades given by the admin.

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14. Advantages of DBMS

The main advantages are:

- Easy data management
- Better security
- Reduced redundancy
- Improved consistency
- Fast data retrieval
- Backup and recovery
- Data sharing
- Better organization

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15. Conclusion

A DBMS provides an organized way to store and manage data. It allows applications to efficiently perform operations such as **insert, select, update, and delete**.

For a project like StudySphere, a database is required to manage student accounts, notes, assignments, submissions, and grades.

⭐ Quick Revision

DBMS = Database Management System

Database → Stores data

Table → Stores records

Primary Key → Uniquely identifies a record

Foreign Key → Connects tables

SQL → Communicates with the database